Design of Experiments Final Report

Aaheli Das

Group Number: 3

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Professor: Dr. Yun Wang

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Executive Summary

This project signifies the importance of looking into microfluidic paper-based analytical devices (µPADs) as they are very widely used for various medical applications. One of the major issues when it comes to creating the µPADs is making the barrier that prevents the fluid from escaping the designated channel. Given that materials used to make the barriers are not all the same and would have different mechanical and chemical properties, the experiment was going to determine whether there would be a significant difference in terms of wicking speed and the leakage area when using different barriers such as; wax, marker, and pencil. To test this the team used a stencil to draw the borders with those barrier materials onto chromatography paper. Then they placed two droplets of 20 microliters of distilled water while taking a video of each run to analyze the wicking speed and leakage area later. Due to repeated discussion with each person in the team, the group decided all together that there will be a few changes to the procedure due to lack of time and to be more efficient with the resources. The team conducted 9 runs for each barrier type and 3 runs for each set of positive and negative controls. After conducting the experiment, it was observed that the pencil had the least leakage area whereas the marker had the highest wicking speed. It was also observed that the difference in performance was statistically significant across all bariers. This goes to show how the experiment rejects the null hypothesis for both conditions since it was proven that the materials made an impact on the outcome of the results.

Significance

It is important to look into paper based microfluidics and creating barriers for them as it closely mimics what is typically used for at-home COVID testing as well as many other applications. Focussing and conducting more research on this topic can help with the reproducibility of gathering reliable results and prevent any sort of false positives and negatives that can lead to a potential misdiagnosis. In addition to that, working with paper based microfluidics is a major game changer due to its low cost, making any product using this technique more affordable for patients as a result making healthcare more accessible.

Objective

The objective of conducting this experiment is to understand what types of barrier would work best to form a microfluidic channel. The following barriers that were being tested were; wax, marker, and pencil. Based on the conditions the type of experiment being conducted was going to be a comparative design where the μPADs were an independent variable.

The team also incorporated a positive control where they doubled the amount of marker being used as the barrier by going over it multiple times. As for the negative control, the team also performed the experiment on chromatography paper with no barriers.

The hypothesis that is going to be focussed on is whether the barrier type affects the performance in terms of leakage area and wicking speed. The null hypothesis stated that there was no difference while the alternative tried to prove that there is a difference. The following official hypotheses are listed below.

H_0: Barrier type does not affect the wicking speed or leakage area.

H_a: Barrier type does affect the wicking speed or leakage area.

I. Materials

For this project, all the group was given a budget of $100 and were told to select the materials that they would need to conduct the experiment. The team had selected the following materials which also include the personal protective equipment that was used to make sure there was no source of contamination and to minimize any risk of errors;

- Chromatography paper (Whatman grade)
- Wax crayons
- Sharpie
- HB or 2B graphite pencil
- ImageJ software
- Laptop or computer
- Food dye
- Distilled water
- Gloves

II. Methods

Started off by placing the stencil on top of the chromatography paper to trace the barriers with either the marker, crayon, or pencil. Then the paper was taken and placed on a red mat that had an outline of the perimeter of the paper and point in the middle of it to help the operator know where to place the droplet of distilled water. For the operator to see the tracing of the outline on the red mat through the paper, the mat had to be placed on top of a torch. The circular chromatography paper was 55 millimeters in diameter and was not changed or cut into different shapes as mentioned previously in the proposal. All the operators were wearing gloves to avoid any sort of contamination.

While a group member added two consecutive droplets of 20 microliters of water that was dyed blue, the other team member recorded a video of the process until the water stopped spreading. The time taken for the 40 microliters of water to spread was recorded to help calculate the wicking speed. The amount of distilled water inserted into each coupon was supposed to originally be 10 microliters but that was changed since it was not enough to reach the barriers.

Another change that was made to the original proposal was getting rid of the hot plate to soften and melt the wax crayons so that it can be poured into the barrier slot. Instead the team decided to make the width of the barriers a little bit wider so that the tips of the crayons can fit and be used to shade in the barrier slot.

Each one of the videos were screenshotted three consecutive times; once two seconds after the first droplet, then two seconds after the second droplet, and lastly one right after the water completely stopped spreading. After these screenshots were taken, the team measured the area of how much the water spread in each image using ImageJ. To do so they set the scale correctly so that ImageJ knows what the measurement is relative to, therefore the team set the distance of the barrier to 25 mm. The team also took measurements of the leakage area past the rectangle formed by the barriers as well as the area of the entire paper. All of these measurements were then used to calculate the leakage area in percent form and the wicking speed.

After the experiment was completed it was time to calculate and analyze the leakage area and wicking speed. The One-way ANOVA is first run to see if the difference in means between the factors is statistically significant or not and if it is significant then Tukey HSD is run to see if differences between each pair of factors are statistically significant or not.

Results

I. Leakage Area Analysis

SUMMARY				
Groups	*Count*	*Sum*	*Average*	*Variance*
Wax	9	456.93	50.77	248.67
Marker	9	580.26	64.47	212.32
Pencil	9	245.35	27.26	123.55

***Table 1**: Basic statistics of Leakage Area data.*

ANOVA						
Source of Variation	*SS*	*df*	*MS*	*F*	*P-value*	*F crit*
Between Groups	6375.84	2	3187.92	16.36	3.29E-05	3.40
Within Groups	4676.35	24	194.85			
Total	11052.19	26				

***Table 2**: One-way ANOVA analysis for Leakage Area.*

Tukey HSD		
MSE	194.85	
qcrit	3.51	
SE	4.65	
HSD	16.31	
Comparison	Difference	
Wax vs Marker	13.70	Not significant
Wax vs Pencil	23.51	Significant
Marker vs Pencil	37.21	Significant

***Table 3**: Significance comparison using Tukey HSD for Leakage Area.*

II. Wicking Speed Analysis

SUMMARY				
Groups	*Count*	*Sum*	*Average*	*Variance*
Wax	9	199.98	22.22	65.93

Marker	9	134.10	14.90	7.05
Pencil	9	153.55	17.06	8.84

***Table 4**: Basic statistics of Wicking Speed data.*

ANOVA						
Source of Variation	*SS*	*df*	*MS*	*F*	*P-value*	*F crit*
Between Groups	254.60	2	127.30	4.67	1.94E-02	3.40
Within Groups	654.56	24	27.27			
Total	909.16	26				

***Table 5**: One-way ANOVA analysis for Wicking Speed.*

Tukey HSD		
MSE	27.27	
qcrit	3.51	
SE	1.74	
HSD	6.10	
Comparison	Difference	
Wax vs Marker	7.32	Significant
Wax vs Pencil	5.16	Not significant
Marker vs Pencil	2.16	Not significant

***Table 6**: Significance comparison using Tukey HSD for Wicking Speed.*

Discussion0

I. Leakage Area

The Leakage area was calculated by taking the difference between the total area and the designed area and then dividing that difference by the area design itself and then multiplying it by 100% to get the value in a percent form.

$$Leakage\ Area\ \% = \frac{Total\ Area\ (mm) - Area\ Design\ (mm)}{Area\ Design\ (mm)} * 100\%$$

***Equation 1:** Leakage area percentage equation.*

Given that the F-statistic value was 16.36 making it higher than the F-crit value of 3.40. There is a statistically significant difference in leakage area between the barrier types so the null

hypothesis is rejected. When using the Tukey HSD comparison, it gives the audience a better grasp of how they compare with one another. It was proven that wax and marker barriers are statistically similar since the difference between them had a value of 13.70 which was lower than the HSD value of 16.31. When comparing the other pairs, the analysis showed something different, it showed how both differences were higher than the HSD meaning that the differences between wax vs pencil and marker vs pencil were statistically different which leads to the conclusion of rejecting the null hypothesis once again.

II. Wicking Speed

The higher the wicking speed, the lower the resistance which indicates that the barrier is weak. To calculate the wicking speed the three areas were summed up and then divided by time that the droplet took to spread. As mentioned before the three areas were calculated as the following points; once two seconds after the first droplet, then two seconds after the second droplet, and lastly one right after the water completely stopped spreading. Here is the equation:

$$Wicking\ Speed\ = \frac{Area1 + Area2 + Area3}{Time}$$

Equation 2: *Wicking speed equation.*

Given that the F-statistic value was 4.67 making it higher than the F-crit value of 3.40. There is a statistically significant difference in wicking speed between at least two of the barrier types so the null hypothesis is rejected. In addition to that, according to the Tukey HSD comparison, wax is significantly faster than marker by a huge margin since it had a greater difference than the HSD value of 6.10 so the null hypothesis is rejected once again. However when wax was compared to pencil there was not a significant difference since it was lower than the HSD value. The same could be said when comparing marker and pencil since their difference was lower than the HSD once again.

III. Observations

One of the aims was to see which materials would perform the best, meaning that it would have the lowest wicking speed and the lowest leakage area. However this was difficult to figure that out since the materials performed differently for each responsible variable. For instance, pencil performed the best for the leakage area percentage however it had the highest

wicking speed making it not the best contender. But according to the results, markers had the lowest wicking speed going to show that it has potential to be a good barrier as well.

Even though statistics showed pencil had the highest leakage area, it was quite apparent from the naked eye that the leakage area for markers seemed to be a lot less than by the other barriers by a huge margin. The markers showed how they were about to block the liquid from barriers so the leakage area for the markers was only composed of the fluid that spread around the barrier, not through it. This could have been due to improper assessment of how the leakage area should be calculated since it didn't account for where it was leaking.

IV. Sources of Error

Due to the fact that the wax crayon was white in color, it was hard to determine where the barriers were which made it harder to set the scale as well as gather the measurements so even though wax performed significantly better in certain conditions it is still important to take this into consideration.

The reason the leakage area might have been higher than expected but still relatively low when the marker was used as a barier. Due to the liquid from the marker having a low viscosity it is easier for the paper to lose its density in the area where the barrier was drawn on. To prevent this in the future, it would be best to only stoke the marker once instead of multiple times.

One of the other reasons the leakage area was higher in marker was because the area of leakage that was used to calculate it was not an ideal way to measure it since the fluid stopped spreading once it hit the barrier but when it was away from the barrier it spread out more.

Another source of error is the fact there were several coupons that were confused for marker or pencil since the team thought that some coupons with pencil seemed a little too similar in resemblence.

Other sources of error were presented when gathering the measurements. For example, the chromatography paper would lose its flatness once the liquid was poured into the middle of it making it difficult to measure the total area of the paper through imageJ.

Conclusion

In conclusion the team decided to reject the null hypothesis since there was enough analysis showing that the performance of the barriers were statistically different when compared

to one another. Hopefully with further testing of each one of the three materials, researchers can get closer to determining what would be the most ideal material to use as a barrier for μPADs used in take home medical tests. The pencil performed better in terms of having the lowest leakage area whereas the marker had the lowest wicking speed making it a good contender for being a barrier. In the future, it would be best to work on an experiment that tests for different characteristics in the same conditions for marker and wax since those had the highest results in different categories. Other than that the process of designing an experiment with a set of constraints was difficult to comprehend at first but was ultimately very helpful since it helped everybody in the team stay on track as well as understand the goals and objectives of the experiment.

References

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Appendix

ChipID	Barrier Type	Sample Condition	Replicate	Sheet Batch	Run Day	Random Order	Operator	Time (s)	Avg. Wicking Speed (mm^2/s)	Avg. Leakage Area %
C001	Wax	Flow (Dye)	1	S1	Day 1	17	Chris	23	11.313	28.359
C002	Marker	Flow (Dye)	1	S1	Day1	42	Bella	21	15.144	31.076
C003	Pencil - HB	Flow (Dye)	1	S1	Day 1	5	Briseyda	21	10.317	80.286
C004	Wax	Flow (Dye)	2	S1	Day 1	33	Chris	25	10.263	41.561
C005	Marker	Flow (Dye)	2	S1	Day 1	2	Rana	19	11.164	38.411
C006	Pencil	Flow (Dye)	2	S1	Day 1	55	Aaheli	22	15.303	42.178
C007	Wax	Flow (Dye)	3	S1	Day 1	51	Chris	24	17.799	42.771

C008	Marker	Flow (Dye)	3	S1	Day 1	31	Briseyda	22	14.128	45.704
C009	Pencil	Flow (Dye)	3	S1	Day 1	47	Bella	21	17.864	67.043
PC1	Positive Control	Wax double	1	S1	Day 1	11	Daniel	20	17.163	20.638
NC1	Negative Control	No Barrier	1	S1	Day 1	9	Daniel	19	14.293	39.083
C010	Wax	Flow (Dye)	4	S2	Day 2	62	Briseyda		25.195	37.820
C011	Marker	Flow (Dye)	4	S2	Day 2	6	Briseyda	22	19.577	37.442
C012	Pencil	Flow (Dye)	4	S2	Day 2	50	Bella	18	19.276	67.156
C013	Wax	Flow (Dye)	5	S2	Day 2	29	Daniel	20	20.887	51.766
C014	Marker	Flow (Dye)	5	S2	Day 2	39	Briseyda	21	14.477	20.089
C018	Pencil	Flow (Dye)	5	S2	Day 2	3	Aaheli	20	15.346	39.199
C019	Wax	Flow (Dye)	6	S2	Day 2	44	Aaheli	18	22.522	72.779
C020	Marker	Flow (Dye)	6	S2	Day 2	19	Rana	19	13.321	16.375
C021	Pencil	Flow (Dye)	6	S2	Day 2	26	Briseyda	21	18.480	62.268
PC2	Positive Control	Wax double	2	S2	Day 2	20	Bella	20	12.481	14.260
NC2	Negative Control	No Barrier	2	S2	Day 2	14	Briseyda	22	13.912	34.829
C022	Wax	Flow (Dye)	7	S3	Day 3	24	Briseyda	21	27.014	49.964
C023	Marker	Flow (Dye)	7	S3	Day 3	31	Aaheli		18.728	21.809
C024	Pencil	Flow (Dye)	7	S3	Day 3	87	Rana	20	19.559	71.959
C025	Wax	Flow (Dye)	8	S3	Day 3	1	Chris	18	31.792	55.388
C026	Marker	Flow (Dye)	8	S3	Day 3	10	Rana	19	13.867	18.705
C027	Pencil	Flow (Dye)	8	S3	Day 3	38	Bella	18	18.863	77.605
C028	Wax	Flow (Dye)	9	S3	Day 3	67	Aaheli	18	33.195	76.524
C029	Marker	Flow (Dye)	9	S3	Day 3	72	Bella	20	13.693	15.735
C030	Pencil	Flow (Dye)	9	S3	Day 3	8	Aahei	19	18.537	72.568
PC3	Positive Control	Wax double	3	S3	Day 3	63	Rana	19	14.971	24.770
NC3	Negative Control	No Barrier	3	S3	Day 3	92	Bella	20	13.462	35.475

Table 7 *: DOE table explaining all the results, conditions and orders of all the runs.*

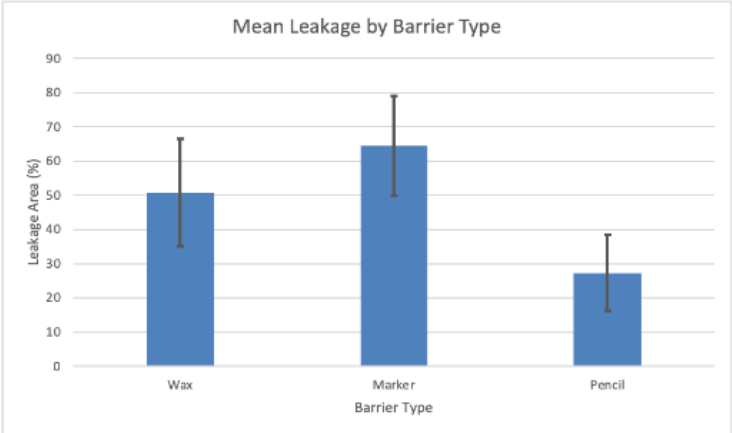

Figure 1: *Mean distribution of leakage area.*

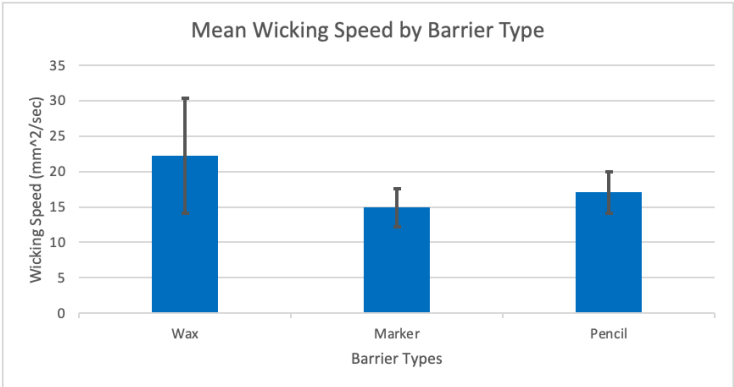

Figure 2: *Mean distribution of wicking speed.*